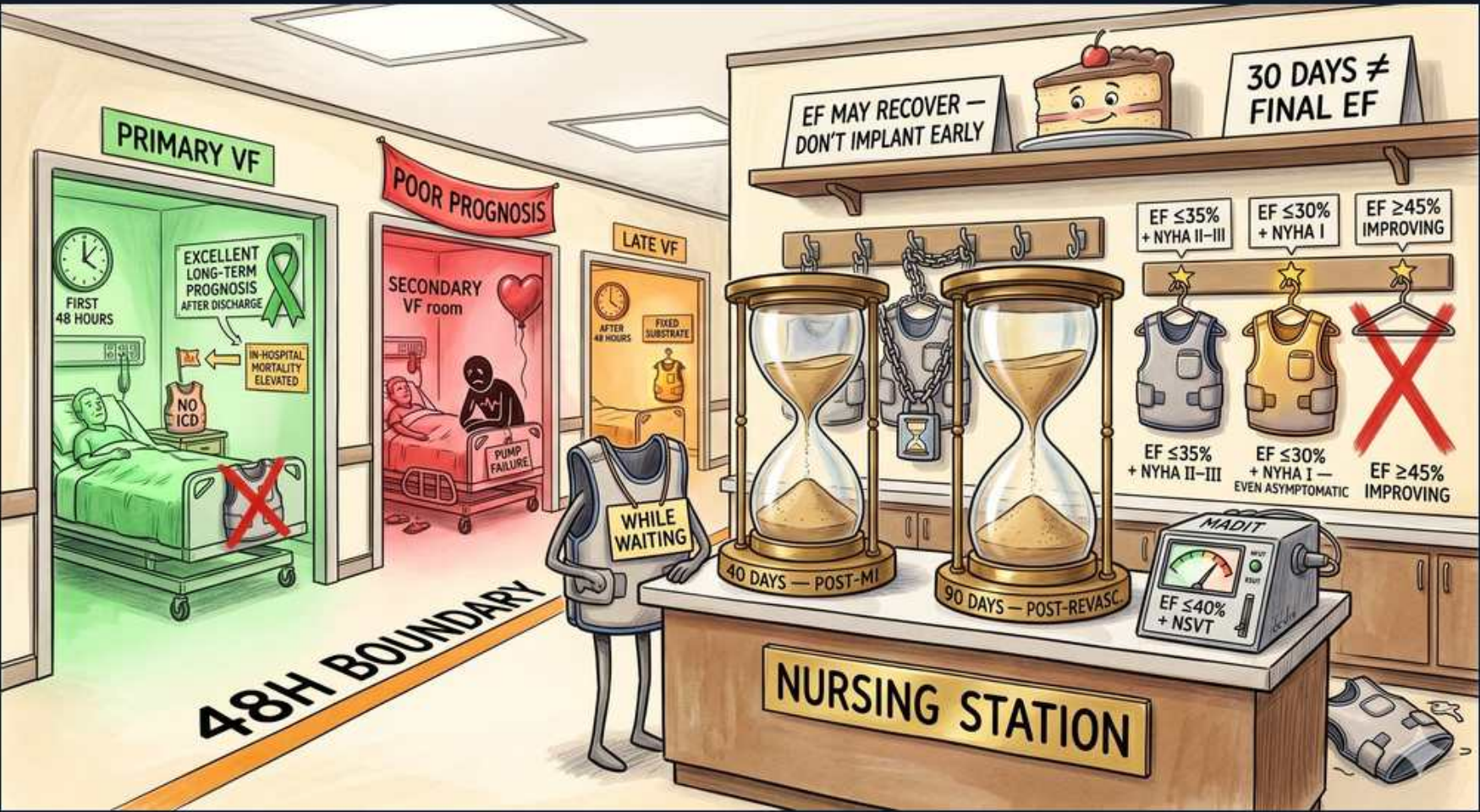


IHD / STEMI — arrhythmia, VF & ICD timing



1. Primary VF room

WHAT YOU SEE

Green doorway 'PRIMARY VF'

WHAT IT ENCODES

Primary VF is ventricular fibrillation occurring in the first 48h of an MI from acute electrical instability of ischemic myocardium (re-entry around the infarct border zone, catecholamine surge, hypokalemia/hypomagnesemia), NOT from pump failure or shock. It is the most common cause of pre-hospital sudden cardiac death in MI. Treat immediately with unsynchronized defibrillation (200 J biphasic) plus amiodarone 300 mg IV bolus; correct K⁺ to >4.0 and Mg²⁺. Mnemonic: 'primary = pure electrical, prognosis preserved.'

BOARD TRAP

Board says: 'Primary VF reflects extensive pump failure and predicts poor outcome.' WRONG — that describes SECONDARY VF (occurs with HF/cardiogenic shock and marks a large infarct). Discriminator: primary VF happens <48h in a hemodynamically intact heart and does NOT worsen long-term prognosis once the patient is revascularized.

2. First 48 hours

WHAT YOU SEE

Clock 'FIRST 48 HOURS' in green room

WHAT IT ENCODES

The 48-hour window defines 'primary/early' VF. VF or sustained VT in the first 48h after STEMI is attributed to transient ischemia/reperfusion and electrical instability, so it does NOT by itself trigger an ICD. After 48h, ventricular arrhythmia is 'late' and signals a fixed arrhythmogenic substrate (scar re-entry) — that DOES carry prognostic weight and may warrant ICD evaluation. Mnemonic: 'the 48-hour clock resets the prognosis.'

BOARD TRAP

Board says: 'Primary VF occurs after day 2 / late in the hospital course.' WRONG. Discriminator: by definition primary (early) VF is WITHIN 48h; arrhythmia beyond 48h is 'late VF/VT' with a different, worse implication and is the type that argues for an ICD.

3. Excellent long-term prognosis

WHAT YOU SEE

Green ribbon 'EXCELLENT LONG-TERM PROGNOSIS AFTER DISCHARGE'

WHAT IT ENCODES

Patients who survive early primary VF and are successfully revascularized have long-term survival similar to MI patients who never had VF. The risk was confined to the acute peri-infarct period, so once discharged their prognosis is excellent and no ICD is needed for that event alone. This is why you do NOT implant a device for an isolated early VF episode. Mnemonic: 'survive the first 48h → survive long-term.'

BOARD TRAP

Board says: 'A patient who had VF on day 1 of his STEMI needs an ICD before discharge because his long-term survival is poor.' WRONG. Discriminator: early primary VF that is defibrillated and revascularized confers EXCELLENT long-term prognosis — the correct answer is reassess EF in ≥40 days, not implant now.

4. No ICD for primary VF

WHAT YOU SEE

Crossed-out ICD 'NO ICD' in green room

WHAT IT ENCODES

An ICD is NOT indicated for VF/VT occurring within the first 48h of MI. The arrhythmia is a transient ischemic phenomenon, not a marker of a permanent substrate, and trials (DINAMIT, IRIS) showed early ICD implantation post-MI provides no mortality benefit and may harm. ICDs are reserved for sustained VT/VF that is unrelated to a transient/reversible cause, or for primary prevention based on EF measured ≥40 days out. Mnemonic: 'early VF, no ICD.'

BOARD TRAP

Board says: 'Any VF episode during an MI is an automatic, lifelong ICD indication.' WRONG. Discriminator: timing is everything — VF <48h = reversible ischemic trigger (no ICD); sustained VT/VF >48h or low EF measured later = ICD.

5. Secondary VF / poor prognosis

WHAT YOU SEE

Red doorway 'POOR PROGNOSIS', 'SECONDARY VF'

WHAT IT ENCODES

Secondary VF occurs in the setting of heart failure, cardiogenic shock, or hemodynamic collapse and reflects a large infarct with pump failure — not just electrical instability. It carries high in-hospital mortality (the death is driven by the underlying pump failure). Management is defibrillation plus aggressive treatment of the shock/HF (revascularization, inotropes/pressors, mechanical support). Mnemonic: 'secondary = shock-associated, survival suffers.'

BOARD TRAP

Board says: 'Secondary VF carries an excellent prognosis like primary VF.' WRONG. Discriminator: secondary VF is a marker of extensive myocardial damage and HF, so in-hospital mortality is markedly ELEVATED — the opposite of primary VF.

6. In-hospital mortality elevated

WHAT YOU SEE

Red figure 'IN-HOSPITAL MORTALITY ELEVATED'

WHAT IT ENCODES

The excess mortality of secondary VF comes from the large infarct/pump failure it accompanies, not the arrhythmia per se. These patients are typically Killip class III–IV. Even with successful defibrillation, in-hospital death is high because the underlying cardiogenic shock persists. This is why secondary VF is a poor-prognosis marker while primary VF is not. Mnemonic: 'big infarct → secondary VF → big mortality.'

BOARD TRAP

Board says: 'VF does not change in-hospital mortality regardless of when it occurs.' WRONG. Discriminator: secondary VF (with HF/shock) significantly raises in-hospital mortality; primary VF (<48h, intact pump) does not change long-term mortality — the context, not the rhythm name, drives the answer.

7. EF may recover — don't implant early

WHAT YOU SEE

Top banner 'EF MAY RECOVER — DON'T IMPLANT EARLY'

WHAT IT ENCODES

Acutely depressed LV function after MI often improves once stunned/hibernating myocardium recovers following revascularization and optimal GDMT. Implanting an ICD during the index admission based on an acutely low EF over-treats patients whose EF will rise above threshold. Therefore defer the ICD decision and re-measure EF after a waiting period. Mnemonic: 'stunned muscle wakes up — wait before you wire.'

BOARD TRAP

Board says: 'Implant an ICD during the initial STEMI admission because the EF is 30%.' WRONG. Discriminator: an acutely low EF may recover — the guideline-correct answer is to optimize GDMT and reassess EF at ≥ 40 days post-MI (or ≥ 90 days post-revascularization) before deciding on an ICD.

8. 30 days ≠ final EF

WHAT YOU SEE

Banner '30 DAYS ≠ FINAL EF'

WHAT IT ENCODES

EF measured at or before 30 days is not the 'final' EF for ICD decisions. Guidelines require reassessment at ≥ 40 days after the MI, or ≥ 90 days after revascularization (PCI/CABG), because recovery continues over weeks. The later, stabilized EF — combined with NYHA class — determines primary-prevention ICD eligibility. Mnemonic: '40 days for the MI, 90 days for the wire (revasc).'

BOARD TRAP

Board says: 'The EF obtained on day 1–2 of the MI determines ICD eligibility.' WRONG. Discriminator: the qualifying EF must be measured after the recovery window (≥ 40 days post-MI / ≥ 90 days post-revasc), not acutely — using the early EF over-implants devices.

9. 40 days post-MI hourglass

WHAT YOU SEE

Left hourglass '40 DAYS POST-MI'

WHAT IT ENCODES

Primary-prevention ICD evaluation must wait ≥ 40 days after the MI (per MADIT-II/SCD-HeFT-derived guidelines and the DINAMIT/IRIS early-implant null results). This waiting period allows EF recovery and avoids implanting in patients with high early non-arrhythmic mortality. At 40 days, re-measure EF; if EF $\leq 35\%$ with NYHA II–III (or $\leq 30\%$ NYHA I), implant. Mnemonic: 'wait forty days after the MI, then decide.'

BOARD TRAP

Board says: 'Refer for ICD at 7 days post-MI if EF is low.' WRONG. Discriminator: primary-prevention ICD eval is deferred to ≥ 40 days post-MI — implanting at one week showed no benefit (DINAMIT) and is the classic distractor.

10. 90 days post-revasc hourglass

WHAT YOU SEE

Right hourglass '90 DAYS POST-REVASC'

WHAT IT ENCODES

If the patient was revascularized (PCI or CABG), wait ~ 90 days before primary-prevention ICD evaluation, because revascularization itself can improve EF over that period. Re-image EF at ≥ 90 days; implant only if it remains $\leq 35\%$ (NYHA II–III) or $\leq 30\%$ (NYHA I). The two clocks are distinct: 40 days post-MI vs 90 days post-revascularization. Mnemonic: 'revascularized = give it 90 days to revive.'

BOARD TRAP

Board says: 'Implant an ICD immediately after PCI in a patient with EF 30%.' WRONG. Discriminator: post-revascularization you must wait ~ 90 days and re-check EF, since the procedure may raise EF above the ICD threshold — immediate implantation is not guideline-supported.

11. EF $\leq 35\%$ + NYHA II–III vest

WHAT YOU SEE

Gray vest 'EF $\leq 35\%$ + NYHA II–III'

WHAT IT ENCODES

Core primary-prevention ICD criterion (SCD-HeFT): EF $\leq 35\%$ with NYHA class II–III heart failure on optimal GDMT, measured after the appropriate waiting period. The ICD prevents sudden cardiac death from scar-related VT/VF. Patient must have a reasonable expected survival > 1 year with good functional status. Mnemonic: 'thirty-five and class two-three → ICD for me.'

BOARD TRAP

Board says: 'An ICD is only indicated when the patient is NYHA class IV.' WRONG. Discriminator: NYHA IV (non-ambulatory, end-stage) patients are generally NOT candidates for a primary-prevention ICD (consider CRT-D, transplant, or palliation). The sweet spot is NYHA II–III with EF $\leq 35\%$.

12. EF $\leq 30\%$ NYHA I vest

WHAT YOU SEE

Yellow vest 'EF $\leq 30\%$ + NYHA I (even asymptomatic)'

WHAT IT ENCODES

Primary-prevention ICD is ALSO indicated at EF $\leq 30\%$ even in NYHA class I (asymptomatic) patients post-MI (MADIT-II criterion). The lower EF threshold compensates for the absence of symptoms. So two pathways qualify: EF $\leq 35\%$ with NYHA II–III, OR EF $\leq 30\%$ with NYHA I. Mnemonic: 'thirty or under, symptoms or not.'

BOARD TRAP

Board says: 'Asymptomatic (NYHA I) patients never qualify for an ICD.' WRONG. Discriminator: an asymptomatic post-MI patient with EF $\leq 30\%$ (measured ≥ 40 days out) DOES qualify for a primary-prevention ICD per MADIT-II.

13. EF $\geq$ 45% improving (no ICD)

WHAT YOU SEE

Crossed vest 'EF $\geq$ 45% IMPROVING'

WHAT IT ENCODES

If the reassessed EF has recovered above threshold (e.g., $\geq$ 45%, and certainly if normalized), there is NO indication for a primary-prevention ICD — the arrhythmic risk has fallen with the improved function. This is the payoff of waiting 40/90 days: you spare patients an unnecessary device. Reassessment, not the acute nadir EF, governs the decision. Mnemonic: 'EF bounced back → no device.'

BOARD TRAP

Board says: 'Implant the ICD anyway even though repeat EF improved to 50%.' WRONG. Discriminator: once EF recovers above the threshold, the patient no longer meets criteria — the correct action is continued GDMT and surveillance, not implantation.

14. EF $\leq$ 40% + NSVT monitor

WHAT YOU SEE

Monitor 'EF $\leq$ 40% + NSVT' at nursing station

WHAT IT ENCODES

In the intermediate zone — EF 36–40% with non-sustained VT (NSVT) on monitoring — patients do not automatically qualify by EF alone. The classic MUSTT/MADIT-I pathway is an electrophysiology (EP) study: if programmed stimulation induces sustained VT, an ICD is indicated. This risk-stratifies patients who fall between clear-cut thresholds. Mnemonic: 'borderline EF + NSVT → send to the EP lab.'

BOARD TRAP

Board says: 'A patient with EF 38% and runs of NSVT needs no further workup.' WRONG. Discriminator: this intermediate group warrants an EP study; inducible sustained VT then justifies an ICD (MADIT-II/MUSTT). Ignoring NSVT in a patient with reduced EF misses a guided ICD indication.
